

Skill - 14

1. Append Redirection and stderr Redirection

The image shows a terminal window on the left and a code editor on the right. The terminal window displays the execution of a program named `skill14.c`. It shows the compilation of `skill13.c` and `skill14.c`, followed by the execution of `./skill14`. The program output shows a simple redirection shell that can handle append redirection (`>>`) and error redirection (`2>`). The code editor on the right shows the source code of `skill14.c`, which implements a simple shell with a `while` loop that reads input from `stdin` and executes commands using `execvp`. It also handles redirection by opening files and redirecting `stdout` and `stderr` to the appropriate file descriptors.

```
harshika@siri:~$ gedit skill14.c

** (gedit:6454): WARNING **: 03:35:33.896: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found

** (gedit:6454): WARNING **: 03:35:33.897: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill13.c -o skill13
harshika@siri:~$ gcc skill14.c -o skill14
harshika@siri:~$ ./skill14
Simple Redirection Shell - Skill 14
Use >> for append and 2> for error redirection.
Type exit to quit.

myshell> echo First Line >> data.txt
Command completed.
myshell> Second Line >> data.txt
execvp: Permission denied
Command completed.
myshell> exit
Shell exited.
harshika@siri:~$ gedit skill14.c

** (gedit:6678): WARNING **: 03:38:39.204: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found

** (gedit:6678): WARNING **: 03:38:39.205: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
```

```
111  if (argc == 0)
112      exit(0);
113
114      execvp(args[0], args);
115
116      perror("execvp");
117      exit(1);
118  }
119
120  /* Parent closes descriptors */
121  if (append_fd != -1)
122      close(append_fd);
123
124  if (error_fd != -1)
125      close(error_fd);
126
127  waitpid(pid, NULL, 0);
128
129  printf("Command completed.\n");
130 }
131
132 int main()
133 {
134     char input[INPUT_SIZE];
135
136     printf("Simple Redirection Shell - Skill 14\n");
137     printf("Use >> for append and 2> for error redirection.\n");
138     printf("Type exit to quit.\n\n");
139
140     while (1)
141     {
142         printf("myshell> ");
143         fflush(stdout);
144
145         if (fgets(input, sizeof(input), stdin) == NULL)
146             break;
147
148         input[strcspn(input, "\n")] = '\0';
149
150         if (strcmp(input, "exit") == 0)
151         {
152             printf("Shell exited.\n");
153             break;
154         }
155
156         if (strlen(input) == 0)
157             continue;
158
159         execute_command(input);
160     }
161
162     return 0;
163 }
```